

Multilabel Feature Selection Based on Fuzzy Mutual Information and Orthogonal Regression

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Abstract—With the increase of high-dimensional multilabel data, multilabel feature selection (MFS) has received more and more widespread attention. Embedded feature selection methods have been widely studied due to their high efficiency and low computational cost. Fuzzy mutual information, as an effective tool for processing continuous features, is widely used in filter feature selection, which results in many repeated entropy calculations. Most of the existing multilabel embedded feature selection methods are based on least squares regression, which loses a lot of statistical and structural information. To solve the above-mentioned problems, we established an optimization framework based on fuzzy mutual information that considers global correlation to obtain the weight of each feature. Under this framework, many repeated entropy operations are avoided. Then, the weight of each feature is introduced into the orthogonal regression optimization framework as prior knowledge. Finally, two optimization frameworks are comprehensively considered for MFS. Furthermore, considering the characteristics of multilabel data, we extend the proposed method to feature-specific MFS. We conducted sufficient experiments to demonstrate the efficiency of our proposed method.

Index Terms—Feature selection, fuzzy rough sets, multilabel learning, mutual information, optimization objective.

I. INTRODUCTION

IN RECENT years, with the continuous increase of multilabel data, multilabel feature selection (MFS) as an effective multilabel data dimensionality reduction method has attracted more and more attention from researchers [1]. Effective MFS methods can greatly improve the generalization performance of the multilabel learning model and save computing resources [2].

Filter, wrapper, and embedded are the three main MFS methods currently used. The filter methods evaluate the importance of features and are independent to learning algorithms [3], [4]. The wrapper methods generate different feature subsets and

evaluate the generated subsets through learning algorithms [5]. The embedded methods combine feature selection with model training and obtains the feature subset by optimizing the objective function [6]. This article mainly studies the embedded method and also contains the idea of the filter.

There are various correlations within multilabel data, including correlations between features, correlations between features and labels, and correlations between labels. For example, the feature “height” has a large correlation with the feature “weight,” while “height” has a small correlation with the feature “birthday.” The feature “height” has a strong correlation with the label “athlete,” while the feature “birthday” has a small correlation with the label “athlete.” The label “athlete” has a strong correlation with labeling “health,” while the label “athlete” has a small correlation with the label “gender.” Therefore, it is necessary to fully consider the multiple correlations in MFS. Mutual information is an effective tool for measuring correlations between two variables. Many scholars have done a lot of research around mutual information [7]. For example, Peng et al. [8] proposed the minimal-redundancy-maximal-relevance, Yang et al. [9] proposed joint mutual information, and Estévez et al. [10] proposed normalized mutual information. However, classic mutual information cannot be used to handle continuous data. In order to solve this problem, Hu et al. [11] combined the fuzzy framework with mutual information and proposed the concept of fuzzy mutual information. Recently, Dai et al. [12] proposed the concept of fuzzy mutual implication granularity information from a granular viewpoint. Currently, many scholars have applied mutual information to MFS. Doquire et al. [13] used a pruning strategy and the multivariate mutual information combined with a problem transformation, and proposed an MFS method that considers dependencies between labels and between features. Qian et al. [14] proposed a label distribution MFS method based on mutual information, which converts logical labels in multilabel data to label distributions with more supervised information. Lin et al. [15] designed neighborhood mutual information to fit multilabel learning. Xiong et al. [16] used the similarity between instances to convert logical labels in multilabel data into label distribution, and proposed a label distribution MFS algorithm based on the fuzzy mutual information. Liu et al. [17] designed fuzzy mutual information suitable for MFS and considered both label dependence and flow label.

The embedded methods have been widely studied by many scholars in the field of MFS because of their excellent feature selection effect and fast speed [18], [19]. Embedded methods allow the algorithm to decide which features to use,

Manuscript received 19 March 2024; revised 4 May 2024; accepted 5 June 2024. Date of publication 17 June 2024; date of current version 4 September 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62376093 and Grant 61976089, in part by the Major Program of the National Social Science Foundation of China under Grant 20&ZD047, in part by the Natural Science Foundation of Hunan Province under Grant 2021JJ30451, Grant 2022JJ30397, and Grant 2024JJ5275, and in part by the Hunan Provincial Science and Technology Project Foundation under Grant 2018RS3065 and Grant 2018TP1018. Recommended by Associate Editor J. Wang. (Corresponding author: Jianhua Dai.)

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TFUZZ.2024.3415176>, provided by the authors.

Digital Object Identifier 10.1109/TFUZZ.2024.3415176

that is, feature selection and algorithm training are performed simultaneously. The most widely used embedded statistical model is least squares regression [20], [21], which obtains the feature coefficient matrix W by minimizing the sum-of-squares error function. Currently, many MFS methods based on least squares regression have been proposed. For example, Nie et al. [22] proposed robust feature selection, which uses the $l_{2,1}$ norm to sparse the feature coefficient matrix W . Jian et al. [23] proposed multilabel informed feature selection, which considers label correlation and the reduced low-dimensional label matrix. Zhang et al. [18] proposed an optimization framework that considers global global correlation and redundancy for MFS, which comprehensively considers feature correlation, feature redundancy and label correlation. A major shortcoming of least squares regression is that it only considers the error in the ordinate and ignores the error in the abscissa of the original point [24]. Orthogonal regression can preserve more statistical and structural information compared to least squares regression [25], [26]. Orthogonal regression can be viewed as least squares regression restricted to the Stiefel manifold [27]. Orthogonal regression can maintain more discriminant properties and obtain better solutions. Xu et al. [28] used orthogonal regression for feature selection and considered the redundancy between features, but could only handle single-label data. Most of the existing embedded MFS methods are based on least squares regression, which may lose a lot of structural information. At the same time, simply using orthogonal regression for MFS ignores the relevance and redundancy of features and cannot adapt well to the characteristics of each dataset. Directly extending a single-label feature selection method based on orthogonal regression to multiple labels cannot be well adapted to the characteristics of multilabel datasets where each feature corresponds to a different weight for each label.

In this article, we propose a new MFS method based on fuzzy mutual information and orthogonal regression (FMIOR). First, we construct a global optimization framework based on fuzzy mutual information that considers the correlation between features and labels, the correlation between labels, and the redundancy between features. Then, we introduce the feature importance matrix obtained by the first framework into the orthogonal regression model to obtain the feature coefficient matrix. The feature importance matrix and feature coefficient matrix evaluate features from a correlation perspective and a regression perspective, respectively. The importance ranking of features is obtained by considering both perspectives. Furthermore, considering the specificity of labels associated with each feature, we propose a new feature-specific MFS method. To assess the effectiveness of our proposed algorithm, we performed an extensive comparison with nine excellent algorithms for four metrics on two classifiers.

The primary contributions of the article can be summarized in the following manner.

- 1) We combine fuzzy mutual information with the embedded method, and comprehensively consider the correlation from the perspective of features and features, features and labels, labels and labels. Moreover, nonnegative matrix

factorization is introduced to improve the interpretability of the results.

- 2) An MFS framework based on orthogonal regression considering feature weights is proposed. This framework uses the feature importance as prior knowledge.
- 3) A feature-specific MFS algorithm is proposed, and the feature importance ranking is comprehensively considered from two perspectives: correlation and regression.
- 4) We performed experiments on fifteen public datasets and conducted a comparative analysis against nine different MFS algorithms. The outcomes demonstrate that our method can deliver superior classification performance.

The rest of this article is organized as follows. In Section II, we present a review of fundamental concepts of fuzzy rough set and fuzzy mutual information entropy. In Section III, we present the proposed method in detail. In Section IV, the optimization strategy for the proposed framework is proposed. In Section V, we conduct experiments to demonstrate the superiority of the proposed method. Finally, Section VI concludes this article.

II. PRELIMINARY KNOWLEDGE

Some basic knowledge about fuzzy mutual information is introduced in this section, which can be found in [29] and [30].

Definition 1: $\tilde{R}$ denotes a fuzzy T-similarity relation defined over the universe if $\tilde{R}$ exhibits the reflexivity, symmetry, and T-transitivity. Reflexivity means that the similarity $\tilde{R}(e_1, e_1)$ is 1. Symmetry dictates that for any two elements e_1 and e_2 in domain, $\tilde{R}(e_2, e_1) = \tilde{R}(e_1, e_2)$. T-transitivity states that for any elements e_1, e_2 , and e_3 in domain, $T(\tilde{R}(e_2, e_3), \tilde{R}(e_3, e_1)) \leq \tilde{R}(e_2, e_1)$.

Definition 2. ([31]): The matrix representation $RM(\tilde{R})$ for a fuzzy relation $\tilde{R}$ defined over a finite universe $\mathcal{U}$ is expressed as follows:

$$RM(\tilde{R}) = \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1n} \\ s_{21} & s_{22} & \dots & s_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{n1} & s_{n2} & \dots & s_{nn} \end{bmatrix} \quad (1)$$

where $s_{ij} \in [0, 1]$ denotes the similarity between e_i and e_j . The number of elements in the universe $\mathcal{U}$ is n .

Gaussian function [31] is adopted to calculate the similarity between elements. The formula is as follows: $s_{ij} = \exp(-\frac{\|e_i - e_j\|}{2\delta^2})$, where $\|e_i - e_j\|$ is the Euclid distance, and δ represents the variance.

Definition 3: Let $\tilde{R}$ be a fuzzy relation over universe $\mathcal{U}$. The basic knowledge of $\tilde{R}$ about $\mathcal{U}$ in an information system can be represented by $\mathcal{U}/\tilde{R}$, which can be expressed as

$$\mathcal{U}/\tilde{R} = \{[e_i]_{\tilde{R}}\}_{i=1}^m \quad (2)$$

where $[e_i]_{\tilde{R}}$ denotes the fuzzy relation about e_i and $\tilde{R}$. It can be expressed as

$$[e_i]_{\tilde{R}} = \frac{s_{i1}}{e_1} + \frac{s_{i2}}{e_2} \dots + \frac{s_{ik}}{e_k} \dots + \frac{s_{im}}{e_m}. \quad (3)$$

Definition 4: The fuzzy information entropy of a fuzzy T-similarity relation $\tilde{G}$ is as follows:

$$FE(\tilde{G}) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|[e_i]_{\tilde{G}}|}{n}. \quad (4)$$

Fuzzy information entropy measures the uncertainty of features by calculating the similarity between instances.

Definition 5: The fuzzy mutual information of two fuzzy T-similarity relations $\tilde{G}$ and $\tilde{F}$ is defined as

$$FI(\tilde{G}; \tilde{F}) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|[e_i]_{\tilde{G}}| \times |[e_i]_{\tilde{F}}|}{n \times |[e_i]_{\tilde{G}} \cap [e_i]_{\tilde{F}}|}. \quad (5)$$

Fuzzy mutual information can effectively measure the correlation between continuous features.

Definition 6: The normalized fuzzy mutual information of two fuzzy T-similarity relations $\tilde{G}$ and $\tilde{F}$ is defined as

$$NFI(\tilde{G}; \tilde{F}) = \frac{2FI(\tilde{G}; \tilde{F})}{FE(\tilde{G}) + FE(\tilde{F})}. \quad (6)$$

Normalized fuzzy mutual information quantifies the value of fuzzy mutual information to between 0 and 1.

III. PROPOSED FRAMEWORK

We propose a new MFS method that considers feature importance from multiple perspectives in this section.

A. Notations

Matrices are represented by uppercase letters and vectors are represented by lowercase letters. M_i represents the i th row of the matrix, and M_j represents the j th column of the matrix M . $\|M_i\|_2$ represents the Euclidean norm of the i th row vector of matrix M . M_{ij} represents the element in the i th row and j th column of the matrix M . $\text{Tr}(M)$ denotes the trace of M . $X \in \mathbb{R}^{m \times n}$ denotes the input data matrix. $Y \in \mathbb{R}^{k \times n}$ denotes the label matrix. n , m , and k , respectively, represent the number of samples, the number of features, and the number of labels. The Frobenius norm of M is defined as $\|M\|_F$. The calculation formula is as follows: $\|M\|_F = (\sum_{i=1}^m \sum_{j=1}^n M_{ij}^2)^{1/2}$. The important symbols used in this article are summarized in Table I.

B. Correlation and Redundancy Optimization

In multilabel data, there exist the correlation between labels, the correlation between labels and features, and redundant information between features. Comprehensive consideration of these factors helps eliminate irrelevant features and select important features. Fuzzy mutual information is an effective tool that can measure correlation and redundancy. Using fuzzy mutual information, we comprehensively consider correlation and redundancy and propose the objective function.

There are three relationships between features and labels: correlation between features and labels, correlation between labels and labels, and redundancy between features and features. Considering the correlation between features and labels, the

TABLE I
DESCRIPTION OF SYMBOLS USED IN THIS ARTICLE

Notations	Descriptions
n	The number of samples
m	The number of features
k	The number of labels
M_i	The i th row element of the matrix M
M_j	The j th column of the matrix M
M_{ij}	The elements of the i th row and j column of the matrix M
X	The feature data matrix containing all samples and $X \in \mathbb{R}^{m \times n}$
Y	The label data matrix containing all samples and $Y \in \mathbb{R}^{k \times n}$
$\tilde{U}$	The feature correlations matrix and $\tilde{U} \in \mathbb{R}^{m \times k}$
$\tilde{V}$	The feature redundancy matrix and $\tilde{V} \in \mathbb{R}^{m \times m}$
$\tilde{R}$	The label discrimination matrix and $\tilde{R} \in \mathbb{R}^{k \times k}$
C	The feature importance matrix and $C \in \mathbb{R}^{m \times k}$
I_n	The n -order identity matrix
W	The feature coefficient matrix and $W \in \mathbb{R}^{m \times k}$
$\text{Tr}(M)$	The trace of matrix M
$\ M\ _F$	The Frobenius norm of matrix M

objective function is designed as follows:

$$\min_C \|C - \tilde{U}\|_F^2 \text{ s.t. } C \geq 0 \quad (7)$$

where $C \in \mathbb{R}^{m \times k}$ denotes the feature importance matrix. $\tilde{U}$ is a constant value matrix that stores the correlation between features and labels. $U_{ij} \in \tilde{U}$ represents the correlation between the i th feature and the j th label and $U_{ij} = NFI(X_{.i}; Y_{.j})$. Further considering the correlation between features, the objective function is defined as follows:

$$\min_C \|C - \tilde{U}\|_F^2 + \sum_{i=1}^k C_{.i}^T \tilde{V} C_{.i} \text{ s.t. } C \geq 0 \quad (8)$$

where $\tilde{V} \in \mathbb{R}^{m \times m}$ is a constant value symmetric matrix that stores the correlation between features. $V_{ij} \in \tilde{V}$ represents the correlation between the i th feature and the j th feature, which is defined as $V_{ij} = NFI(X_{.i}; X_{.j})$. The second term ensures that if the correlation between two features is high, then the weight of both features in the feature importance matrix C will not be high at the same time. We remove redundant information through the second term.

For two highly correlated labels, the values in their corresponding feature importance matrix C should also be similar. Based on this thought, the objective function is defined as follows:

$$\min_C \|C - \tilde{U}\|_F^2 + \sum_{i=1}^k C_{.i}^T \tilde{V} C_{.i} + \sum_{i=1}^k \sum_{j=1}^k R_{ij} C_{.i}^T C_{.j} \text{ s.t. } C \geq 0 \quad (9)$$

where $R_{ij} = 1 - NFI(Y_i; Y_j)$. R_{ij} represents the degree of distinction between the i th label and the j th label.

Since $\sum_{i=1}^k C_{:,i}^T \tilde{V} C_{:,i} = \text{Tr}(C^T \tilde{V} C)$ and $\sum_{i=1}^k \sum_{j=1}^k R_{ij} C_{:,i}^T C_{:,j} = \text{Tr}(\tilde{R} C^T C)$, the objective function is rewritten as follows:

$$\begin{aligned} \min_C & \|C - \tilde{U}\|_F^2 + \alpha \text{tr}(C^T \tilde{V} C) + \beta \text{tr}(\tilde{R} C^T C) \\ \text{s.t. } & C \geq 0 \end{aligned} \quad (10)$$

where α and β are two tradeoff parameters. The objective function comprehensively considers the correlation between features and labels, labels correlation, and redundancy between features. Different from Zhang et al.'s [18] work, fuzzy mutual information is used to replace mutual information in the proposed objective function. Mutual information is used to process discrete data from the perspective of probability theory. But for continuous data, mutual information will lose a lot of information. Fuzzy mutual information starts from the perspective of similarity between samples and can well retain the distinguishing information between samples. At the same time, the normalized fuzzy mutual information can well ensure that the discrimination between each pair of labels in the label discrimination matrix $\tilde{R}$ is between 0 and 1. Compared with the filter feature selection method based on fuzzy mutual information that has been studied more, the proposed objective function only needs to perform a mutual information operation between pairs of features and pairs of labels, which can avoid a lot of repeated entropy calculations. By optimizing the objective function, we can get the importance of the i th feature according to $\|C_{:,i}\|_2$.

C. Orthogonal Regression Optimization Considering Feature Importance

Compared with least squares regression, orthogonal regression can retain more statistical and structural information. Orthogonal regression can be interpreted as a form of least squares regression that incorporates orthogonal constraints. Based on orthogonal regression and considering the importance of features, the following objective function is proposed:

$$\begin{aligned} \min_{W,b} & \|W^T DX + b1_n^T - Y\|_F^2 \\ \text{s.t. } & W^T W = I_n \end{aligned} \quad (11)$$

where $W \in \mathbb{R}^{m \times k}$ is the feature coefficient matrix that satisfies the orthogonal constraint. $D \in \mathbb{R}^{m \times m}$ is a diagonal matrix. D_{ii} represents the importance of the i th feature and $D_{ii} = (m \times \|C_{:,i}\|_2) / \sum_{j=1}^m \|C_{:,j}\|_2$.

The objective function takes the feature importance calculated in formula (10) as prior knowledge, and minimizes the vertical distance from the data point to the fitting function taking into account the difference in importance of each feature.

IV. OPTIMIZATION STRATEGY

A. Solve C Using Lagrange Multiplier Method

Introducing Lagrange multipliers $\phi \in \mathbb{R}^{m \times k}$ into the objective function to restrict C to be nonnegative, the following

function can be obtained:

$$\begin{aligned} \min_C & \|C - \tilde{U}\|_F^2 + \alpha \text{Tr}(C^T \tilde{V} C) \\ & + \beta \text{Tr}(\tilde{R} C^T C) - \text{Tr}(\phi C^T). \end{aligned} \quad (12)$$

Derivating the above-mentioned formula with respect to C , we get the following formula:

$$\begin{aligned} & \frac{\partial (\|C - \tilde{U}\|_F^2 + \alpha \text{Tr}(C^T \tilde{V} C) + \beta \text{Tr}(\tilde{R} C^T C) - \text{Tr}(\phi C^T))}{\partial C} \\ & = 2(I + \alpha \tilde{V})C + 2\beta C \tilde{R} - 2\tilde{U} - \phi. \end{aligned} \quad (13)$$

Using the Karush–Kuhn–Tucker conditions, the following equation can be obtained:

$$[(I + \alpha \tilde{V})C + \beta C \tilde{R} - \tilde{U}] \circ C = 0. \quad (14)$$

Then, we can get the update rule for C

$$C = C \circ \frac{\tilde{U}}{(I + \alpha \tilde{V})C + \beta C \tilde{R} + \varepsilon} \quad (15)$$

where ε is a small constant to ensure that the denominator is not zero.

B. Solve W by Generalized Power Iteration (GPI)

To solve the least squares regression problem with orthogonal constraints, we transform it into a quadratic problem on the Stiefel manifold (QPSM) and use the GPI algorithm to solve it [32].

We derive formula (11) with respect to b to get

$$\frac{\partial (\|W^T DX + b1_n^T - Y\|_F^2)}{\partial b} = 0. \quad (16)$$

Then, we can obtain

$$b = \frac{1}{n} (Y1_n - W^T DX1_n). \quad (17)$$

Substituting into the above-mentioned formula, formula (11) translates into

$$\begin{aligned} \min_W & \|W^T DXH - YH\|_F^2 \\ \text{s.t. } & W^T W = I_k \end{aligned} \quad (18)$$

where $H = I_n - (1/n)1_n1_n^T$.

The formula (18) can be transformed into the formula form proposed by Nie et al. [32] to solve orthogonal least squares regression. The formula is as follows:

$$\min_{W^T W = I_k} \text{Tr}(W^T ZW - 2W^T K) \quad (19)$$

where $Z = DXHX^T D^T$ and $K = DXHY^T$. The GPI algorithm can well solve the QPSM of the form in formula (19). The solution process of the formula (19) is shown in Algorithm 1.

Algorithm 1: The GPI Algorithm.

Require: the matrix $Z \in \mathbb{R}^{m \times m}$ and matrix $K \in \mathbb{R}^{m \times k}$.
Ensure: the matrix $W \in \mathbb{R}^{m \times k}$.
1: Initialize the parameter v such that $\bar{Z} = vI_m - Z$ is a positive definite matrix;
2: **repeat** :
3: Update $Q = 2\bar{Z}W + 2K$;
4: Calculate $Q = USV^T$ by the compact SVD method;
5: Update $W = UV^T$;
6: **until** convergence
7: **return** W

C. Proposed Algorithm

According to the detailed analysis and theory in the previous sections, we propose a new MFS algorithm based on FMIOR. There are two modules in the proposed algorithm. The first part considers correlation and redundancy to solve the feature importance. The second part considers least squares regression with orthogonal constraints to obtain the feature coefficient matrix. The second part builds on the first part. Finally, the results obtained from the two parts are combined to comprehensively consider the importance of the features. Correspondingly, the algorithm is proposed, as shown in Algorithm 2. The time complexity of FMIOR mainly includes two parts: calculating fuzzy mutual information matrix and executing the optimization framework. The time complexity of calculating fuzzy correlation matrix between features is $O(n^2m^2)$. The time complexity of calculating fuzzy correlation matrix between features and labels is $O(n^2mk)$. The time complexity of calculating fuzzy correlation matrix between labels and labels is $O(n^2k^2)$. The time complexity of the Lagrange multiplier method in the correlation optimization framework and the augmented power iteration in the regression framework is $O(m^2n + m^2k)$. Therefore, the total time complexity of FMIOR depends on the number of features and the number of labels. If the number of features is greater than the number of labels, the total time complexity of FMIOR is $O(n^2m^2)$. If the number of labels is greater than the number of features, the total time complexity of FMIOR is $O(n^2k^2)$.

D. Extension to Feature-Specific Feature Selection

Through Algorithm 2 we get the importance of features for each label. However, not every feature is related to every label. For example, the feature “age” has nothing to do with the label “gender.” Each feature has its specific associated label. Therefore, better performance can be obtained by considering the labels specifically related to each feature when selecting features.

The two matrices W and C obtained in Algorithm 2 can be used as prior knowledge for feature-specific feature selection. Feature-specific feature selection builds the model by searching for the top $k\%$ labels most relevant to each feature. For each feature a_i , we only consider the top $k\%$ labels with a high correlation with the feature, which can be described by the

Algorithm 2: The Feature Selection Algorithm FMIOR.

Require: the feature matrix X , the label matrix Y , parameters α and β .
Ensure: the feature coefficient matrix W and the feature importance matrix C .
1: Initialize W and C randomly, where W satisfies the orthogonality constraint; $t=0$;
2: Compute matrix $\tilde{U}$, $\tilde{V}$ and $\tilde{R}$;
3: **repeat**
4: $C_{ij}^{t+1} \leftarrow C_{ij}^t \frac{\tilde{U}_{ij}}{((I+\alpha\tilde{V})C+\beta C\tilde{R})_{ij}+\epsilon}$;
5: $t=t+1$;
6: **until** convergence
7: Compute D via C ;
8: Compute $Z = DXHX^TD^T$;
9: Compute $K = DXHY^T$;
10: Compute the dominant eigenvalue v of Z ;
11: Compute $\bar{Z} = vI_m - Z$;
12: **repeat** :
13: Update $Q = 2\bar{Z}W + 2K$;
14: Calculate $Q = USV^T$ by the compact SVD method;
15: Update $W = UV^T$;
16: **until** convergence
17: **return** W and C

formula as follows:

$$a_{ij} = \begin{cases} a_{ij}, j \in \text{Top}_{ik} \\ 0, \text{otherwise} \end{cases} \quad (20)$$

where a_{ij} denotes the importance of label j to the feature a_i . Top_{ik} represents the set of top $k\%$ labels that rank the importance of feature a_i .

Specifically, in this article, we calculate the new feature-specific importance matrix C^{new} and feature-specific coefficient matrix W^{new} according to formula (20). Then we get two new feature importance rankings Rank_C and Rank_W by the values of $|C_{i.}^{\text{new}}|_2$ and $|W_{i.}^{\text{new}}|_2$, respectively. Finally, we sort the average ranking of each feature on Rank_C and Rank_W to get the feature importance sequence. Combining the feature-specific idea, a new MFS method is proposed based on FMIOR.

V. EXPERIMENTAL ANALYSIS

In this section, we perform a comprehensive set of experiments to demonstrate the effectiveness of our proposed FMIOR.

A. Experimental Design

In our experimental setup, we employ fifteen publicly available datasets.¹ These datasets have been widely used in multi-label learning. The details of these real datasets are presented in Table II.

We use four evaluation metrics to evaluate the performance of the algorithm, including average precision, Macro-F1, Micro-F1, and coverage. Detailed definitions of these metrics can be

¹[Online]. Available: <http://www.uco.es/kdis/mlresources/>

TABLE II
DESCRIPTION OF DATASETS

Datasets	Samples	Features	Label	Train	Test
Arts	5000	462	26	2000	3000
Business	5000	438	30	2000	3000
Computers	5000	681	33	2000	3000
Education	5000	550	33	2000	3000
Enron	1702	1001	53	1123	579
Entertainment	5000	640	21	2000	3000
Genbase	662	1186	27	463	199
Health	5000	612	32	2000	3000
Image	2000	294	5	1000	1000
PlantGO	978	3091	12	588	390
Recreation	5000	606	22	2000	3000
Reference	5000	793	27	2000	3000
Slashdot	3782	1079	22	1891	1891
Society	5000	636	27	2000	3000
Yeast	2417	103	14	1500	917

found in Wu and Zhou's [33] work. Among these metrics, the larger the value of Macro-F1, Micro-F1, and average precision, the better the effect. The smaller the value of coverage, the better the effect. The multilabel classifiers used in this article are ML-KNN [34] (with default settings) and binary support vector machine (BSVM) [35]. Due to page limit, all experimental results using BSVM are presented in the supplementary material.

To demonstrate the performance of FMOIR, we conduct a comparative analysis with nine effective MFS algorithms, including MDFS [36], GRRO [37], PMU [38], MCLS [39], SCLS [40], MCDM [41], VMFS [42], GLFS [43], and SSFS [44]. A brief introduction to the nine comparison algorithms is as follows.

- 1) *MDFS*: It is a manifold regularized feature selection method that constructs a low-dimensional embedding based on the original feature space, fits the label distribution, and locally captures the label correlation.
- 2) *GRRO*: It is a global optimization framework that takes into account feature correlation, feature redundancy, and label correlation.
- 3) *PMU*: It is a modified model to maximize feature relevance and minimize feature redundancy.
- 4) *MCLS*: It is a method based on manifold-constrained Laplacian score, which comprehensively considers the influence of supervisory information and local features.
- 5) *SCLS*: It is capable of handling large label set using the designed scalable relevance criterion.
- 6) *MCDM*: It is a method based on multicriteria decision-making, which regards MFS as a process of information fusion.
- 7) *VMFS*: It is a MFS method based on multicriteria decision-making and VIKOR.
- 8) *GLFS*: This method performs feature selection by considering both the special characteristics of labels in the same group and the specific characteristics possessed by each label.
- 9) *SSFS*: This method is based on latent structure shared term and uses graph regularization technology to ensure the

consistency between the original feature space and the latent feature structure space.

For our proposed FMIOIR, parameters α and β are searched in $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. The parameter k is searched in $[20, 40, 60, 80, 100]$. For comparison methods with multiple parameters, the parameter settings are as follows. For MDFS, the parameters β and γ are searched in $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. For SSFS, the parameters α , β , and γ are searched in $[0.1, 0.3, 0.5, 0.7, 0.9, 1]$. For GLFS, the parameters α and λ are set to 1, and the parameters β and γ are searched in $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. For GRRO, the parameters α and β are searched in $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. We select the parameters with the highest average precision as the optimal parameters.

B. Experimental Results and Analysis

For fifteen datasets, we take the top 50 feature subsets and calculate their average values on four metrics. The experimental results on the ML-KNN classifier are shown in Tables II–V. Due to space limitations, we put the table of the experimental results on the BSVM classifier in the supplementary material.

For metrics average precision, Macro-F1, and Micro-F1, the larger their values, the better. Tables III–V, respectively, show the experimental results of the average precision, Macro-F1, and Micro-F1 of the ten algorithms on the ML-KNN classifier. From Table III, we can see that FMIOIR achieves the highest average precision on eleven datasets. From Table IV, we can see that FMIOIR achieves the highest Micro-F1 metric on nine datasets. From Table V, we can see that FMIOIR achieves the highest Macro-F1 metric on nine datasets. For metrics coverage, the smaller their values, the better. Table VI shows the experimental results of coverage of the ten algorithms on the ML-KNN classifier. From Table VI, we can see that FMIOIR achieves the lowest Coverage on ten datasets. Overall, our method FMIOIR has superior performance than the other nine methods on the four metrics of the ML-KNN classifier. The experimental results observed on the BSVM classifier are similar.

TABLE III
EXPERIMENTAL RESULTS ON ML-KNN WITH AVERAGE PRECISION $\uparrow$

Datasets	FMIOR	MDFS	GRRO	PMU	MCLS	SCLS	MCDM	VMFS	GLFS	SSFS
Arts	0.5259	0.5218	0.4484	0.4453	0.4301	0.4547	0.4816	0.4512	0.5234	0.4902
Business	0.8747	0.8733	0.8614	0.8594	0.8608	0.8599	0.8625	0.8624	0.8719	0.8695
Computers	0.6274	0.6271	0.6078	0.6088	0.6037	0.6099	0.6081	0.6098	0.6246	0.6127
Education	0.5518	0.5574	0.4931	0.4840	0.4790	0.4840	0.5019	0.4918	0.5593	0.5339
Entertainment	0.5895	0.5871	0.5080	0.5311	0.4917	0.5356	0.4828	0.5290	0.5742	0.5440
Enron	0.5537	0.5487	0.5528	0.5329	0.5256	0.5511	0.5276	0.5256	0.5532	0.5356
Genbase	0.9169	0.9269	0.9163	0.9124	0.4782	0.9214	0.9167	0.4783	0.9421	0.9246
Health	0.7236	0.7226	0.6308	0.6136	0.6136	0.6216	0.6109	0.6301	0.7205	0.6708
Image	0.4222	0.4207	0.3808	0.4074	0.3708	0.4069	0.3895	0.3984	0.4213	0.3964
PlantGO	0.8274	0.8456	0.8483	0.8416	0.6968	0.8384	0.5487	0.4988	0.8420	0.8100
Recreation	0.5370	0.5267	0.4144	0.4222	0.3924	0.4357	0.4060	0.3912	0.5280	0.4658
Reference	0.6378	0.6336	0.5752	0.5848	0.5793	0.5895	0.5682	0.5834	0.6267	0.6073
Slashdot	0.6749	0.5279	0.5343	0.4979	0.3548	0.5290	0.4259	0.3548	0.5283	0.5053
Society	0.5962	0.5886	0.5499	0.5498	0.5516	0.5482	0.5495	0.5546	0.5916	0.5801
Yeast	0.7496	0.7526	0.7496	0.7361	0.7481	0.7350	0.7319	0.7383	0.7512	0.7411
Win	11	1	1	0	0	0	0	0	2	0

TABLE IV
EXPERIMENTAL RESULTS ON ML-KNN WITH MICRO-F1 $\uparrow$

Datasets	FMIOR	MDFS	GRRO	PMU	MCLS	SCLS	MCDM	VMFS	GLFS	SSFS
Arts	0.2213	0.2154	0.0249	0.0487	0.0013	0.0663	0.1668	0.0148	0.2356	0.1022
Business	0.6925	0.6916	0.6677	0.6673	0.6677	0.6675	0.6841	0.6685	0.6913	0.6830
Computers	0.3161	0.4086	0.3492	0.3771	0.3724	0.3633	0.2864	0.3315	0.3936	0.3465
Education	0.2256	0.2381	0.0391	0.0159	0.0085	0.0185	0.1393	0.0082	0.2619	0.1542
Entertainment	0.3337	0.3022	0.0894	0.1294	0.0545	0.1222	0.1672	0.1259	0.2896	0.1832
Enron	0.3973	0.3729	0.3402	0.3340	0.3065	0.3559	0.3263	0.3064	0.3432	0.3207
Genbase	0.7758	0.7644	0.8219	0.7769	0.1694	0.8538	0.7946	0.1694	0.8274	0.8175
Health	0.5293	0.5290	0.3451	0.2922	0.2933	0.2473	0.2331	0.2769	0.5230	0.3893
Image	0.1430	0.1362	0.0942	0.1332	0.0754	0.1314	0.0988	0.0885	0.1418	0.1118
PlantGO	0.6743	0.7265	0.7349	0.7171	0.5036	0.7089	0.2017	0.0241	0.7174	0.6684
Recreation	0.2729	0.2621	0.0608	0.0894	0.0190	0.1211	0.0931	0.0096	0.2716	0.1474
Reference	0.3487	0.3964	0.2441	0.2802	0.2099	0.2986	0.3453	0.2746	0.3905	0.3357
Slashdot	0.4300	0.3344	0.3416	0.2730	0.0061	0.3377	0.1677	0.0061	0.3283	0.2889
Society	0.2935	0.2670	0.1561	0.1677	0.1759	0.1778	0.1705	0.2279	0.2834	0.2585
Yeast	0.6144	0.6128	0.6119	0.5878	0.5979	0.5840	0.5770	0.5986	0.5996	0.6135
Win	9	2	1	0	0	1	0	0	2	0

TABLE V
EXPERIMENTAL RESULTS ON ML-KNN WITH MACRO-F1 $\uparrow$

Datasets	FMIOR	MDFS	GRRO	PMU	MCLS	SCLS	MCDM	VMFS	GLFS	SSFS
Arts	0.0937	0.0843	0.0098	0.0168	0.0006	0.0222	0.0512	0.0088	0.0887	0.0447
Business	0.1338	0.1603	0.0990	0.0981	0.0995	0.0975	0.1173	0.1011	0.1520	0.1350
Computers	0.0935	0.0723	0.0313	0.0325	0.0301	0.0302	0.0324	0.0303	0.0511	0.0414
Education	0.1299	0.1248	0.0759	0.0687	0.0653	0.0682	0.0897	0.0728	0.1285	0.1100
Entertainment	0.1509	0.1369	0.0349	0.0683	0.0158	0.0610	0.0485	0.0553	0.1180	0.0869
Enron	0.0602	0.0718	0.0471	0.0516	0.0395	0.0642	0.0519	0.0395	0.0759	0.0473
Genbase	0.4528	0.3759	0.4379	0.3537	0.0327	0.4469	0.4170	0.0327	0.4464	0.4334
Health	0.2176	0.2265	0.1233	0.0868	0.0834	0.0958	0.0849	0.1173	0.2181	0.1783
Image	0.1428	0.1345	0.1060	0.1298	0.0836	0.1276	0.1139	0.1020	0.1417	0.1202
PlantGO	0.4806	0.5035	0.5331	0.4701	0.2415	0.4708	0.0809	0.0118	0.4889	0.3673
Recreation	0.1399	0.1322	0.0331	0.0382	0.0135	0.0518	0.0257	0.0077	0.1348	0.0777
Reference	0.1115	0.1354	0.0800	0.0958	0.0806	0.0996	0.0819	0.0851	0.1278	0.1067
Slashdot	0.1010	0.1630	0.1527	0.1112	0.0024	0.1493	0.0553	0.0023	0.1510	0.1211
Society	0.0784	0.0696	0.0162	0.0164	0.0182	0.0189	0.0172	0.0225	0.0695	0.0572
Yeast	0.3319	0.3303	0.3287	0.2916	0.3101	0.2893	0.2736	0.3076	0.3311	0.3085
Win	9	4	1	0	0	0	0	0	1	0

TABLE VI
EXPERIMENTAL RESULTS ON ML-KNN WITH COVERAGE ↓

Datasets	FMIOR	MDFS	GRRO	PMU	MCLS	SCLS	MCDM	VMFS	GLFS	SSFS
Arts	5.3940	5.4783	5.9238	5.9338	6.0881	5.8984	5.9150	5.8593	5.5637	5.5572
Business	2.3049	2.3562	2.5343	2.6054	2.5789	2.5980	2.5488	2.5344	2.4085	2.4492
Computers	4.2815	4.4426	4.7644	4.8107	4.7519	4.8371	4.7609	4.7470	4.5909	4.6717
Education	3.9938	3.9371	4.3466	4.4730	4.5519	4.4519	4.3174	4.3491	4.0072	4.0465
Entertainment	3.2452	3.2610	3.6981	3.4984	3.8444	3.5080	3.9402	3.5236	3.3744	3.4174
Enron	16.0254	16.0299	16.3025	16.7096	16.8658	16.1476	16.5085	16.8658	16.0258	16.4153
Genbase	0.8401	1.4597	1.5487	1.4781	4.7607	1.4216	1.5499	4.7607	1.2704	1.5371
Health	3.1002	3.2235	3.9236	4.0697	4.0842	3.9612	4.1205	3.8263	3.3114	3.4946
Image	2.6847	2.6707	2.9438	2.7859	3.0290	2.7762	2.8954	2.7736	2.6315	2.8974
PlantGO	0.7620	0.7046	0.7004	0.7054	1.4541	0.7134	2.5201	2.7117	0.7311	0.9141
Recreation	4.6079	4.7602	5.4302	5.3971	5.4748	5.3653	5.5746	5.5133	4.6834	5.1320
Reference	3.3435	3.4193	3.8512	3.8186	3.8099	3.7852	3.9255	3.7043	3.5250	3.5657
Slashdot	3.8157	3.8781	3.8081	4.1142	5.1861	3.8541	4.7868	5.1861	3.8542	4.0726
Society	5.6657	5.7903	6.1904	6.1772	6.1669	6.1923	6.1782	6.0933	5.7513	5.8866
Yeast	6.4937	6.4845	6.4836	6.6347	6.5384	6.6425	6.6937	6.5945	6.4783	6.5919
Win	10	1	2	0	0	0	0	0	2	0

In order to more specifically demonstrate the experimental results, we use line charts to depict the value changes of each algorithm on average precision and coverage as the number of selected features changes. Due to space limitations, we display figures of the classification performance on the ML-KNN and BSVM classifiers as a function of the number of selected features in the supplementary material. Different algorithms are represented by lines of different colors and shapes. As can be seen from the figures, FMIOR has good performance and robustness on various datasets. Our proposed FMIOR outperforms the other nine methods as the number of selected features changes.

The main reasons why the experimental results of FMIOR are better than other comparison algorithms are as follows. Compared with PMU, which is a modification method based on maximum correlation and minimum redundancy, FMIOR takes global correlation and redundancy into account. Compared with GRRO using mutual information to measure correlation, FMIOR use fuzzy mutual information, which can process continuous features more effectively. Therefore, our method performs better than GRRO, especially on continuous data. SCLS can handle large label data, but its accuracy is lower than FMIOR. MCDM and VMFS are methods based on multicriteria decision-making. They can quickly sort features according to their importance. MCDM and VMFS run faster than FMIOR, but its accuracy is much lower than FMIOR. Compared with MDFS and GLFS based on least squares regression, FMIOR retains more structural information based on orthogonal regression. SSFS retains the latent feature structure and latent label structure, while FMIOR adopts fuzzy information theory to capture the relationship between features and labels. FMIOR is more computationally expensive but has better performance than SSFS. MCLS uses manifold learning to transform logical label space into Euclidean label space, but does not take correlation and redundancy into account well. Compared with MCLS, FMIOR comprehensively considers the correlation and redundancy between features, as well as the correlation between labels, so FMIOR performs better. Compared with other comparison MFS methods, FMIOR takes more comprehensive considerations into MFS. It not only considers feature-label correlation, label-label correlation and feature-feature redundancy,

it also retains more structural information through orthogonal regression.

C. Convergence and Complexity Analysis

We further show the convergence of our proposed FMIOR framework through experiments. The FMIOR framework consists of two parts: the correlation and redundancy optimization framework and the orthogonal regression optimization framework. Figs. 1 and 2, respectively, show the changes in the target values of the two optimization frameworks as the number of iterations increases. It is obvious that the function value decreases rapidly and stabilizes after several iterations. It is obvious that the function value decreases rapidly and stabilizes after several iterations, which demonstrates the convergence of FMIOR.

Furthermore, we compare the running time of our proposed methods and comparison methods. Since the correlation matrix is specific for each dataset and only needs to be calculated once, for FMIOR and GRRO, we input the correlation matrix as prior knowledge. The computation time of the second line of Algorithm 2 is not discussed here. The running time of our proposed methods and comparison methods is shown in Table VII. MCDM and VMFS, two MFS methods based on multicriteria decision-making, are faster than methods based on information theory metrics. The GRRO method converges quickly and therefore has a short running time. The running time of MDFS, MCLS, SCLS, and GLFS are of the same order of magnitude. Compared with other methods, our method does not have an advantage in running time. This is due to the fact that our method evaluates the importance of features from multiple aspects, increasing the computational effort. Although this increases the running time, it makes our method have better performance on various evaluation metrics.

D. Parameter Analysis

There are three parameters in FMIOR, where α and β measure the contribution of relevant and redundant terms, and k determines the number of most relevant labels for each feature when measuring feature importance. We conduct experiments to analyze the role of each parameter in feature selection.

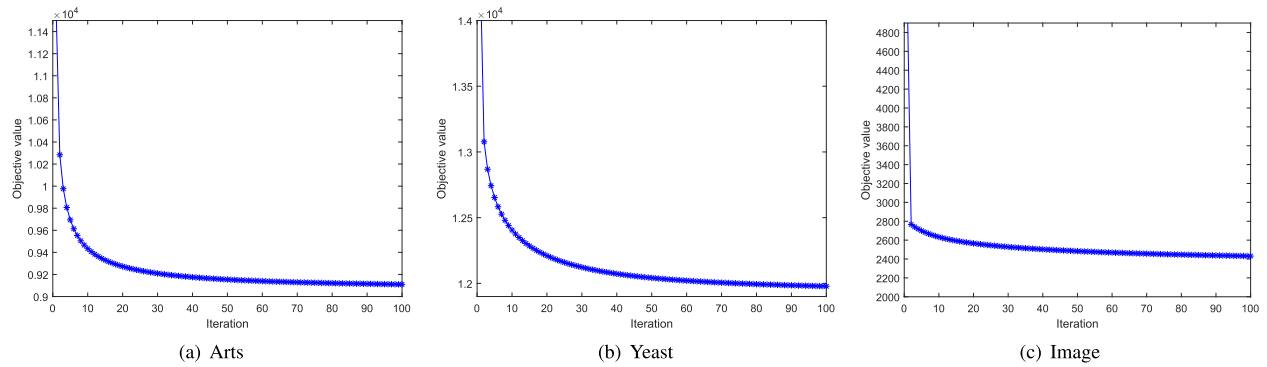


Fig. 1. Objective value of the regression optimization framework changes with the number of iterations. (a) Arts. (b) Yeast. (c) Image.

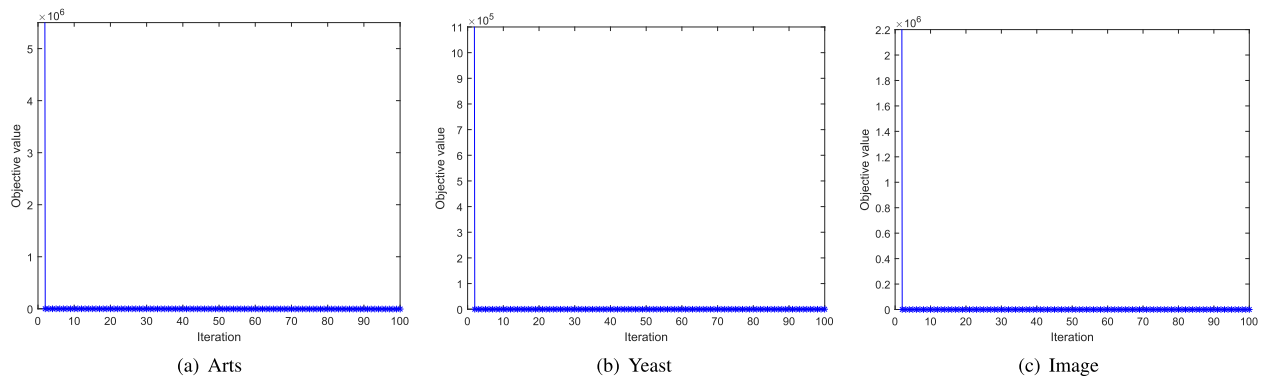


Fig. 2. Objective value of the correlation optimization framework changes with the number of iterations. (a) Arts. (b) Yeast. (c) Image.

TABLE VII
RUNNING TIME (SECONDS) COMPARISON OF TEN METHODS

Datasets	FMIOR	MDFS	GRRO	PMU	MCLS	SCLS	MCDM	VMFS	GLFS	SSFS
Arts	1.75	26.62	0.13	1565.62	23.79	33.85	0.14	1.20	48.75	0.56
Business	1.69	25.28	0.02	1808.95	22.59	32.03	0.08	0.32	27.75	0.51
Computers	2.80	36.25	0.14	3210.12	24.64	50.26	0.06	0.85	58.45	0.88
Education	3.16	27.20	0.08	3062.96	34.51	39.73	0.06	0.26	44.12	0.68
Enron	11.95	21.37	1.81	4524.06	11.86	69.13	0.14	0.50	55.77	0.44
Entertainment	3.61	27.77	0.08	2207.93	22.03	39.53	0.08	0.28	11.50	0.80
Genbase	1.41	3.00	1.17	335.64	2.94	4.91	0.20	0.19	14.00	0.28
Health	3.94	22.27	0.08	3094.68	22.26	43.04	0.06	0.29	7.75	0.51
PlantGO	22.58	17.91	29.83	531.85	6.78	18.98	0.86	0.36	81.68	1.50
Recreation	2.23	33.64	0.13	1841.78	22.06	37.95	0.06	0.19	8.26	0.59
Reference	2.06	30.39	0.20	3794.60	26.51	57.09	0.13	0.38	10.96	0.81
Slashdot	0.50	38.64	2.56	2377.18	38.48	73.37	0.50	0.58	46.17	1.07
Society	1.38	28.53	0.02	2472.84	23.25	42.95	0.06	0.22	9.29	0.57

First, the influence of α and β on FMIOR is analyzed. The parameter k is fixed to 100. We focus on analyzing the impact of parameters change on the average precision. Fig. 3 shows how the accuracy changes as α and β change on arts, yeast, and image. The accuracy is the average of the average precision of the first 50 feature subsets. The figures generated by other datasets are similar to those observed on the arts, image, and yeast. For the dataset arts, we can find that the accuracy decreases when the parameter α takes a larger value or a smaller value, and as the value of the parameter β increases, the accuracy gradually

increases. For the dataset yeast, we can find that the accuracy decreases as the value of parameter α increases, and the accuracy is not sensitive to parameter β . The figure of the image has roughly the same pattern as the figure of yeast. From Fig. 3, we can see that when parameters α and β take intermediate values, FMIOR works better. This proves that the two terms following parameters α and β are equally important.

Then, the parameter k of the feature-specific feature selection method is analyzed. The parameters α and β are fixed to 1. We focus on analyzing the impact of the parameter

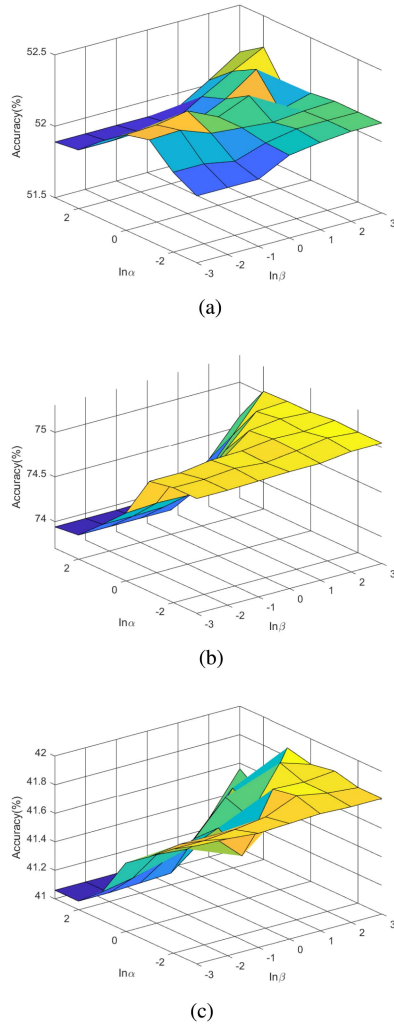


Fig. 3. Changes in average precision as parameters α and β change. (a) Arts. (b) Yeast. (c) Image.

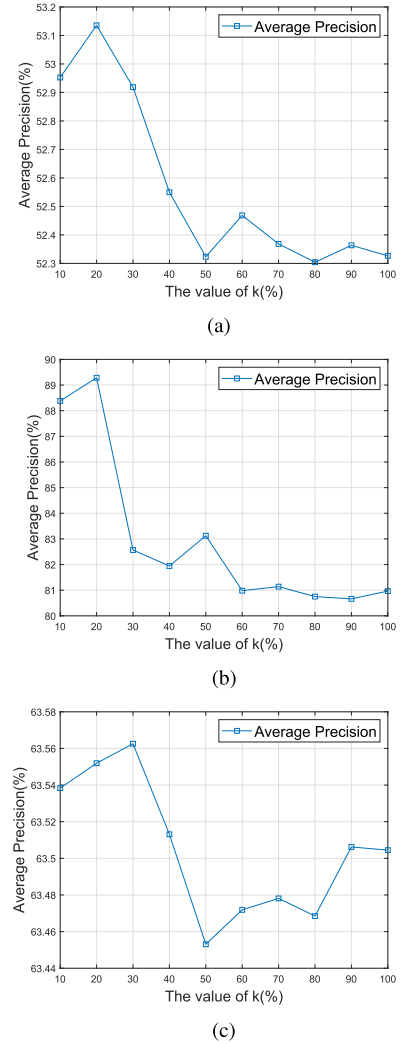


Fig. 4. Changes in average precision as parameter k changes. (a) Enron. (b) Genbase. (c) Reference.

change on the average precision. Fig. 4 shows how the accuracy changes as k changes on Enron, Genbase, and Reference. From Fig. 4, we can see that average precision tends to decrease as the number of labels considered increases. Average precision is highest when considering the top twenty or thirty percent of labels associated with each feature. This phenomenon shows that the labels associated with each feature are often only partial, so we only need to consider some labels that are highly associated with each feature when evaluating feature importance. From Fig. 4, we can see that the classification effect when considering all labels is worse than when considering the most relevant part of the labels. This demonstrates the effectiveness of the feature-specific feature selection method. Based on the experimental results, we recommend that parameter k takes a value within 50.

E. Ablation Studies

Fuzzy mutual information, correlation and redundancy optimization and orthogonal regression are important components of FMIOR. They all play an important role in our proposed algorithm. Here, we use ablation studies to prove the effectiveness

of these three parts for FMIOR. We designed three degraded versions of FMIOR. To demonstrate the impact of fuzzy mutual information on FMIOR, we designed FMIOR-1. FMIOR-1 represents the FMIOR method that uses mutual information instead of fuzzy mutual information. To demonstrate the impact of correlation and redundancy optimization on FMIOR, we designed FMIOR-2. FMIOR-2 represents the FMIOR method without correlation and redundancy optimization. To demonstrate the impact of orthogonal regression optimization on FMIOR, we designed FMIOR-3. FMIOR-3 represents the FMIOR method without orthogonal regression optimization. We use ML-KNN as the classifier. We compared FMIOR and its three degraded versions in terms of coverage and average precision on image, yeast, and genbase. The experimental results are shown in Figs. 5 and 6. From Fig. 5, we can see that FMIOR outperforms the other three degraded versions on all datasets according to the coverage. From Fig. 6, we can see that FMIOR has the highest average precision on all datasets compared to the other three degraded versions. The results of ablation experiments show that fuzzy mutual information, correlation redundancy optimization

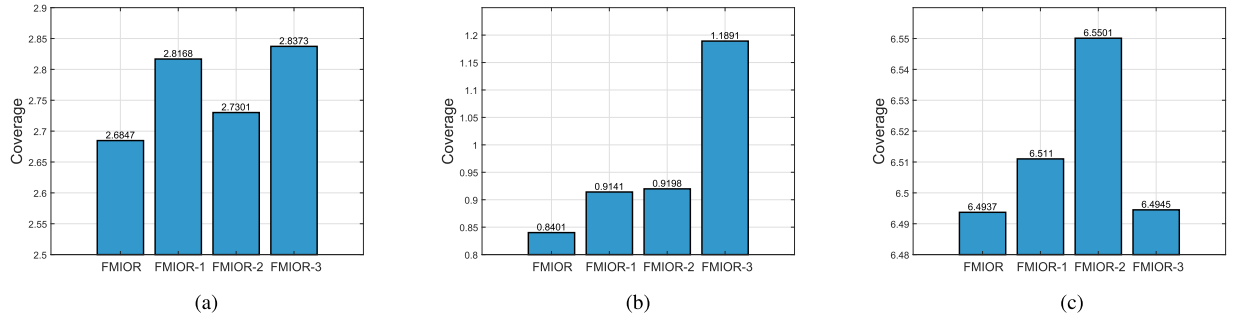


Fig. 5. Comparing the coverage of the proposed method and its variants. (a) Image. (b) Genbase. (c) Yeast.

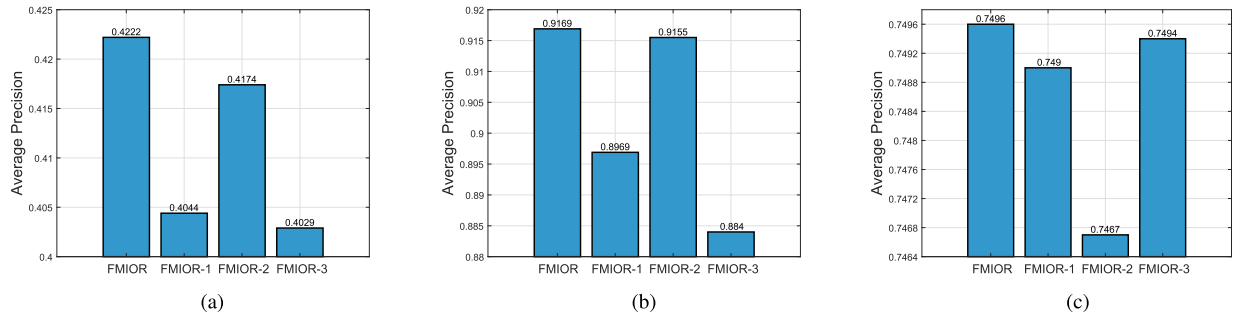


Fig. 6. Comparing the average precision of the proposed method and its variants. (a) Image. (b) Genbase. (c) Yeast.

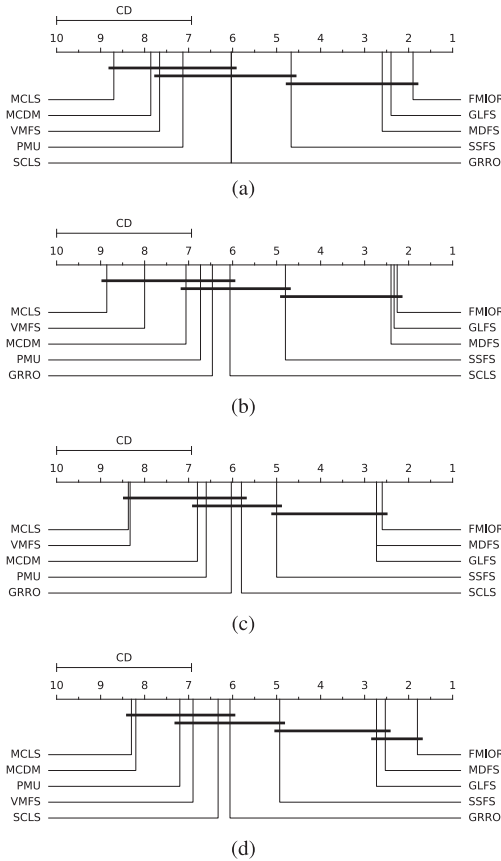


Fig. 7. Performance comparison with ten algorithms on four metrics with the Bonferroni–Dunn test. (a) Average precision. (b) Macro-F1. (c) Micro-F1. (d) Coverage.

and orthogonal regression all play a positive role in MFS and are indispensable.

F. Statistical Significance Test

To demonstrate that there are significant differences among the ten algorithms, we performed the Bonferroni–Dunn test [45] and the Friedman test [46]. ML-KNN is used as the classifier. The Friedman test can determine the existence of significant differences among ten algorithms. From Tables III–VI, it is obvious that the null hypothesis of Friedman’s test is rejected when $\alpha = 0.05$. This means that there are significant differences between the eight algorithms.

Subsequently, the Bonferroni–Dunn post hoc test is used to evaluate the performance differences among the ten algorithms. The formula of the critical distance is

$$CD_{\alpha} = q_{\alpha} \sqrt{\frac{n(n+1)}{6m}} \quad (21)$$

where $q_{\alpha} = 2.773$ with $\alpha = 0.05$ can be found in Demšar’s [47] work.

We get $CD_{0.05} = 3.06$ by calculating Formula (21) when $m = 15$ and $n = 10$. Fig. 7 shows the CD diagrams for ten algorithms on four metrics. When the difference between the average rankings of the two algorithms is greater than the CD value, there is a significant difference between them. Through Fig. 7, we can see that FMIOR is significantly better than PMU, SCLS, GRRO, MCDM, VMFS, and MCLS in terms of average precision. FMIOR is significantly better than PMU, SCLS, GRRO, MCDM, VMFS, and MCLS in terms of Macro-F1. FMIOR is significantly better than PMU, SCLS, GRRO,

MCDM, VMFS, and MCLS in terms of Micro-F1. FMIOR is significantly better than PMU, SCLS, GRRO, MCDM, VMFS, and MCLS in terms of coverage. Our proposed FMIOR algorithm significantly outperforms most comparison algorithms on four metrics. Through the Bonferroni–Dunn test and Friedman test, it is proved that there is a significant difference between our algorithm and the comparison algorithms. Overall, our method has the best performance among the ten algorithms on four metrics.

VI. CONCLUSION

In this article, an MFS method based on FMIOR is proposed. First, we comprehensively consider global correlation based on fuzzy mutual information and propose an optimization framework. Subsequently, utilizing the global correlation optimization framework as prior knowledge, an orthogonal regression-based MFS optimization framework considering feature weights is proposed. Finally, both optimization frameworks are considered together and extended to feature-specific feature selection. Experimental results show that our proposed method achieves good classification performance on various metrics. In the future, we will further study the application of orthogonal regression in MFS. At the same time, we also consider applying the feature-specific method to multilabel classification.

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